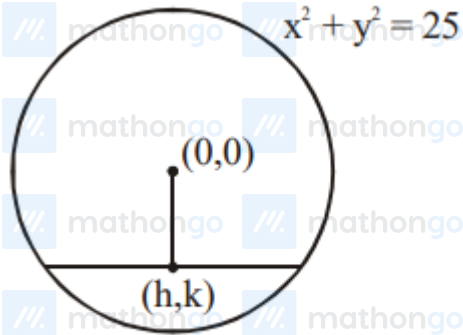


Q1 (4)



Equation of chord

Equation of chord

$$y - k = -\frac{h}{k}(x - h)$$

$$ky - k^2 = -hx + h^2$$

$$hx + ky = h^2 + k^2$$

$$y = -\frac{hx}{k} + \frac{h^2 + k^2}{k}$$

$$\text{tangent to } \frac{x^2}{9} - \frac{y^2}{16} = 1$$

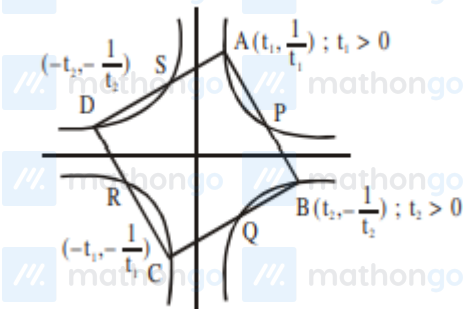
$$c^2 = a^2 m^2 - b^2$$

$$\left(\frac{h^2 + k^2}{k}\right)^2 = 9\left(-\frac{h}{k}\right)^2 - 16$$

$$(x^2 + y^2)^2 = 9x^2 - 16y^2$$

Q2 (80)

$$xy = 1, -1$$



Hints and Solutions

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$$\frac{t_1+t_2}{2} \cdot \frac{\frac{1}{t_1}-\frac{1}{t_2}}{2} = 1$$

$$\Rightarrow t_1^2 - t_2^2 = 4t_1t_2$$

$$\frac{1}{t_1^2} \times \left(-\frac{1}{t_2^2}\right) = -1 \Rightarrow t_1t_2 = 1$$

$$\Rightarrow (t_1t_2)^2 = 1 \Rightarrow t_1t_2 = 1$$

$$t_1^2 - t_2^2 = 4$$

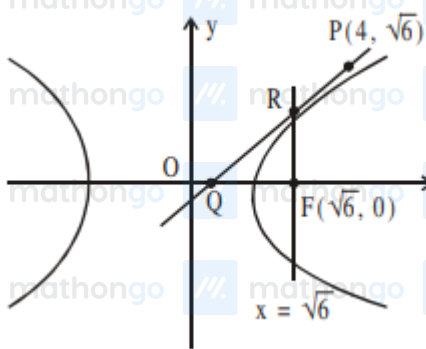
$$\Rightarrow t_1^2 + t_2^2 = \sqrt{4^2 + 4} = 2\sqrt{5}$$

$$\Rightarrow t_1^2 = 2 + \sqrt{5} \Rightarrow \frac{1}{t_1^2} = \sqrt{5} - 2$$

$$AB^2 = (t_1 - t_2)^2 + \left(\frac{1}{t_1} + \frac{1}{t_2}\right)^2$$

$$= 2 \left(t_1^2 + \frac{1}{t_1^2}\right) = 4\sqrt{5} \Rightarrow \text{Area}^2 = 80$$

Q3 (3)



$$\frac{x^2}{4} - \frac{y^2}{2} = 1$$

$$e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{\frac{3}{2}}$$

$$\text{Focus } F(ae, 0) \Rightarrow F(\sqrt{6}, 0)$$

equation of tangent at P to the hyperbola is

$$2x - y\sqrt{6} = 2$$

tangent meet x-axis at Q(1, 0)

$$\& \text{ latus rectum } x = \sqrt{6} \text{ at } R\left(\sqrt{6}, \frac{2}{\sqrt{6}}(\sqrt{6}-1)\right)$$

$$\therefore \text{Area of } \Delta_{QFR} = \frac{1}{2}(\sqrt{6}-1) \cdot \frac{2}{\sqrt{6}}(\sqrt{6}-1)$$

$$= \frac{7}{\sqrt{6}} - 2$$



Q1: 16 March (Shift 1) - Single Correct

The locus of the midpoints of the chord of the circle, $x^2 + y^2 = 25$ which is tangent to the hyperbola,

$\frac{x^2}{9} - \frac{y^2}{16} = 1$ is :

(1) $(x^2 + y^2)^2 - 16x^2 + 9y^2 = 0$

(2) $(x^2 + y^2)^2 - 9x^2 + 144y^2 = 0$

(3) $(x^2 + y^2)^2 - 9x^2 - 16y^2 = 0$

(4) $(x^2 + y^2)^2 - 9x^2 + 16y^2 = 0$

Q2: 18 March (Shift 1) - Numerical

A square ABCD has all its vertices on the curve $x^2y^2 = 1$. The midpoints of its sides also lie on the same curve. Then, the square of area of ABCD is

Q3: 18 March (Shift 2) - Single Correct

Consider a hyperbola $H : x^2 - 2y^2 = 4$. Let the tangent at a point $P(4, \sqrt{6})$ meet the x-axis at Q and latus rectum at R (x_1, y_1) , $x_1 > 0$. If F is a focus of H which is nearer to the point P, then the area of $\triangle QFR$ is equal to

(1) $4\sqrt{6}$

(2) $\sqrt{6} - 1$

(3) $\frac{7}{\sqrt{6}} - 2$

(4) $4\sqrt{6} - 1$

Answer Key

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Q1 (4)

Q2 (80)

Q3 (3)

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